

REVIEW

Baroreceptor Mechanotransduction: Diverse Sensors, Unified Signals

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ABSTRACT: Arterial baroreceptors constitute the safeguard of cardiovascular homeostasis by translating vascular stretch into reflex control/ afferent firing. Recent advances have identified an unexpectedly diverse palette of candidate ion channels, including PIEZO1/2, TMEM150C, TRPV1, TRPC5 (transient receptor potential canonical 5), and epithelial sodium channels (ENaC), yet the molecular logic of their mechanosensory function remains unresolved. However, the field is mired in controversy: are PIEZOs truly indispensable barosensors, or do TRPs, ENaCs, and Tentonin3 provide parallel, context-dependent mechanotransduction? Equally contentious is the extent to which distinct baroreceptor morphologies, end-net versus flower-spray terminals, rapidly versus slowly adapting afferents, deploy specialized channel repertoires to encode dynamic versus steady-state pressure stimuli. Such heterogeneity challenges prevailing reductionist models and underscores baroreception as an emergent property of distributed molecular systems rather than a single-channel solution. Here, we critically appraise molecular, anatomic, and physiological evidence to advance an unifying framework in which baroreceptor diversity is both structural and functional, shaped by channel composition, terminal architecture, and afferent excitability.

Key Words: arterial pressure ■ fibrosis ■ heart rate ■ homeostasis ■ hypertension

Baroreception is a fundamental homeostatic mechanism that enables the cardiovascular system to maintain arterial pressure through the continuous sensing of vascular wall stretch and the modulation of autonomic output.^{1,2} Baroreceptors, located primarily in the carotid sinus and aortic arch, transduce mechanical deformation into neural signals conveyed by the glossopharyngeal and vagus nerves to the nucleus tractus solitarius, where central autonomic circuits regulate heart rate, vascular resistance, and cardiac contractility.^{3,4} Dysfunction of this feedback loop plays a pivotal role in cardiovascular disease pathogenesis. Impaired baroreflex sensitivity arising from defective mechanotransduction, altered afferent excitability, or disrupted central integration drives chronic sympathetic overactivation, parasympathetic withdrawal, and maladaptive cardiovascular remodeling. Such impairment may stem from arterial stiffness and adventitial remodeling, which hinder effective transmission of mechanical strain to sensory nerve endings, as well as from fibrosis, extracellular matrix alterations, and changes in peripheral glial cells.

This dysregulation exacerbates hypertension, heart failure, arrhythmias, and sudden cardiac death.^{5–9}

In heart failure, blunted baroreceptor signaling sustains excessive sympathetic drive, worsening ventricular remodeling and impairing output. Similarly, in resistant hypertension, impaired baroreceptor feedback perpetuates pressure elevation despite intensive pharmacological therapy. Therapeutic strategies such as baroreceptor activation therapy¹⁰ and renal denervation^{11,12} aim to reestablish autonomic balance but remain constrained by an incomplete understanding of the underlying molecular mechanisms.

Recent advances in sensory neurobiology have revealed that barosensation, the detection of blood pressure changes, emerges from a complex interplay of mechanosensitive ion channels. Candidates include Piezo1, Piezo2,¹³ Tentonin3 (TMEM150C),¹⁴ TRPC5,¹⁵ TRPV1,^{16,17} ENaCs,^{18,19} and ASICs (acid-sensing ion channels).²⁰ Yet, fundamental uncertainties remain. Although Piezo channels are frequently highlighted as dominant mediators, other studies implicate Tentonin3,

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TRPs, and ENaCs, suggesting that mechanotransduction is not governed by a single molecular entity but distributed across multiple pathways. Baroreceptors themselves are not uniform entities but a heterogeneous population distinguished by terminal architecture, innervation territory, and firing dynamics. End-net versus flower-spray terminals,^{21–24} carotid versus aortic afferents,^{15,25} and rapidly versus slowly adapting fibers²⁶ may rely on distinct mechanosensory channel assemblies to encode dynamic versus steady-state pressure. This organization is consistent with the integration of high-sensitivity, phasic A-fiber input and low-sensitivity, tonic C-fiber input, enabling both rapid buffering of pressure perturbations and stable long-term regulation of mean arterial pressure, as demonstrated in Seagard's integrative studies and subsequent work. Moreover, the existence of a single baroreceptor unit comprising multiple sensors²⁷ points to baroreception as an integrated, multimodal sensory system²⁸ rather than a uniform pressure gauge subtype-specific repertoire of mechanosensory machinery.

This complex anatomic and functional diversity indicates that baroreception is a composite property, arising from the integration of direct mechanotransduction, chemo-mechanosensory modulation, and excitability-dependent signaling both across distinct and within baroreceptor populations.²⁹ Such diversity challenges reductionist single-sensor models toward an integrative framework that views baroreception as an emergent, multimodal system.

This review critically appraises molecular, anatomic, and physiological evidence to advance a unifying framework in which baroreceptor diversity is both structural and functional, shaped by channel composition, terminal architecture, and afferent excitability. We highlight methodological limitations and potential confounding effects from pharmacological and genetic approaches that contribute to ongoing interpretive uncertainty. By linking ion channel biology to systems-level dysfunction, we propose that targeted modulation of specific baroreceptor subtypes may reprogram reflex arcs and restore cardiovascular balance.

Ultimately, resolving the molecular code of barosensation could transform cardiovascular therapeutics. We argue that the next era of baroreflex research must pivot from single-molecule attribution to integrative, multimodal frameworks that place mechanosensory diversity at the core of blood pressure regulation. Notably, ohmic data obtained in humans suggest that mechanically activated ion channels with properties consistent with those identified in experimental models also play a central role in human baroreception,³⁰ thereby strengthening the translational relevance of findings derived from laboratory animals. Such a shift will not only reconcile long-standing controversies but also establish a foundation for precision interventions capable of reshaping autonomic control at its most fundamental sensory nodes.

Baroreflex in Acute and Long-Term Control of Arterial Pressure

Arterial baroreceptors play a central role in both the short- and long-term regulation of arterial pressure through integrated neural and neurohumoral mechanisms. In the short term, baroreceptor activation provides a rapid negative feedback loop in which increases in arterial pressure enhance afferent firing from the carotid sinus and aortic arch, leading to inhibition of central sympathetic outflow and reciprocal activation of parasympathetic pathways, thereby reducing heart rate, cardiac contractility, and peripheral vascular resistance (Figure 1). This reflex operates on a beat-to-beat basis and is essential for buffering acute perturbations in blood pressure, such as those induced by postural changes or hemorrhage.

Historically, baroreceptors were thought to have little influence on long-term blood pressure regulation due to their ability to reset to prevailing pressure levels.³¹ However, evidence indicates that this resetting is incomplete and that sustained baroreceptor activation can chronically suppress sympathetic nerve activity.³² In particular, prolonged activation of the baroreflex induces durable reductions in arterial pressure through persistent inhibition of sympathetic outflow, especially to the kidney, thereby modulating renal sodium handling and extracellular fluid volume.³¹ Additionally, baroreflex activation interacts with key hormonal systems, including suppression of the renin-angiotensin-aldosterone system, further contributing to long-term pressure control.³³ Further, contemporary therapeutic approaches, such as chronic baroreflex activation and renal denervation, provide strong functional evidence for a long-term role of the baroreflex by demonstrating sustained reductions in arterial pressure associated with decreased central sympathetic tone.³¹ Together, these findings support a revised paradigm in which baroreceptors are not only acute stabilizers of blood pressure but also an integrated regulator contributing to the long-term control of arterial pressure, in close interaction with renal and neurohumoral mechanisms.

Conversion of the Arterial Blood Pressure Waveform Into Neural Firing

The neural code itself is multiplexed, combining rate and temporal coding. Mean arterial pressure is primarily encoded by changes in average firing frequency, while the pulsatile component of blood pressure is encoded by phase-locked bursting synchronized with the cardiac cycle (eg, systolic bursting).³⁴ At intermediate pressures, fibers exhibit systolic phase-locked bursts, whereas at higher pressures they transition to more continuous firing, reflecting nonlinear dynamics of excitability and spike generation. Nonlinear neuronal dynamics further shape encoding, including threshold effects, adaptation,

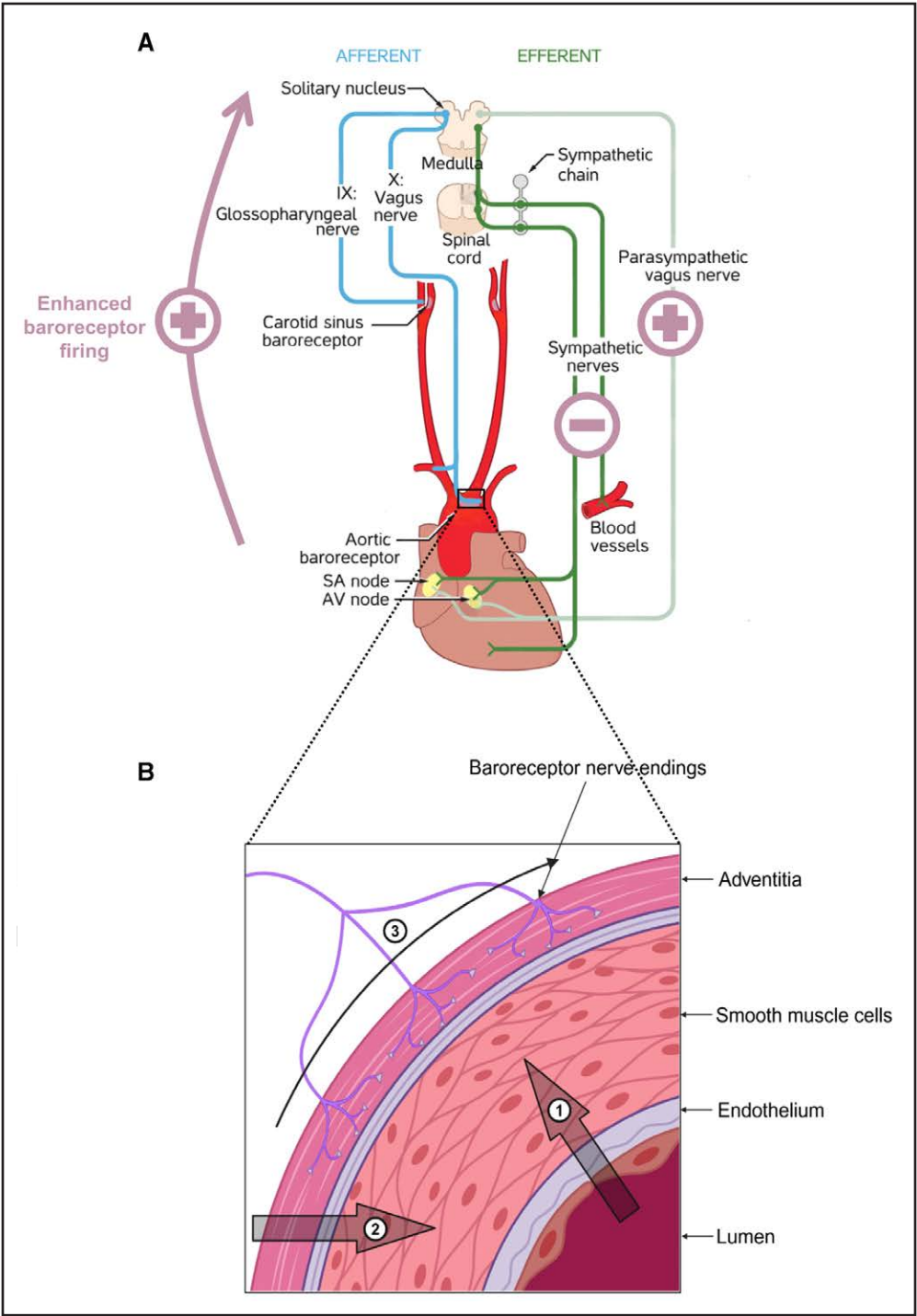


Figure 1. Baroreceptor-mediated neural regulation of arterial pressure.

A, Diagram of the baroreflex circuitry showing the detection of arterial pressure changes by baroreceptors in the carotid sinus and aortic arch and the transmission of afferent signals via the petrosal and nodose ganglia to the central cardiovascular regulatory center in the brainstem. Enhanced baroreceptor firing increases parasympathetic activity and suppresses sympathetic outflow, leading to a reduction in heart rate and arterial pressure, whereas diminished afferent input elicits the opposite response. This negative-feedback mechanism provides rapid buffering of acute fluctuations in arterial pressure. **B**, Location and innervation of arterial baroreceptors. Baroreceptor nerve endings sense circumferential wall stretch (labeled 3), which results from transmural pressure. Transmural pressure is the difference between blood pressure (intraluminal pressure, labeled 1) pushing outward on the aortic wall and external pressure (extramural pressure, labeled 2) pushing inward. The greater the net outward force, the greater the circumferential stretch detected by the baroreceptors. Created with BioRender.com.

and bifurcation-like transitions between firing modes. Computational and experimental studies show that noise, membrane conductances, and dynamic input fluctuations

interact to produce diverse firing patterns (eg, regular spiking, bursting, or intermittent firing), allowing the system to encode both steady-state and rapidly changing

signals.³⁴ Together, these processes transform a continuous, pulsatile mechanical signal into a structured neural output that combines firing rate, spike timing, and fiber-type-specific contributions, providing the central nervous system with a high-fidelity representation of both instantaneous and mean arterial pressure.

Aortic Versus Carotid Sinus Baroreceptors

Accumulating evidence indicates that baroreceptors located in the carotid sinus and aortic arch are not functionally redundant but exhibit distinct mechanosensory and reflex properties. At the peripheral level, differences in intrinsic mechanosensitivity have been directly demonstrated. Using complementary *in vitro* and *in vivo* approaches, Lau et al⁴ showed that aortic baroreceptor neurons display lower activation thresholds and enhanced pressure-evoked firing compared with carotid baroreceptor neurons. These findings were corroborated by larger stretch-induced intracellular Ca^{2+} responses and higher afferent discharge in the aortic depressor nerve, indicating that aortic baroreceptors may be intrinsically more sensitive to mechanical stimuli than their carotid counterparts.⁴

Beyond peripheral encoding, these differences translate into distinct contributions to central baroreflex pathways. Functional dissection of baroafferent inputs *in situ* revealed a dominant role for aortic baroreceptors in mediating cardiac parasympathetic responses. Selective denervation experiments demonstrated that baroreflex bradycardia is critically dependent on aortic afferents, with minimal contribution from carotid sinus inputs, whereas sympathoinhibition can be elicited from either aortic or carotid sites.³⁵ Notably, carotid sinus stimulation, even at high pressures, produced robust reductions in sympathetic activity without significant changes in heart rate, underscoring a dissociation between reflex limbs depending on afferent origin.³⁵ These observations align with the concept of functional specialization within central integration circuits, particularly in the nucleus tractus solitarius, where anatomically distinct afferent pathways, via the glossopharyngeal nerve for carotid inputs and the vagus nerve for aortic inputs, maintain partial segregation.³⁶ Such an organization supports differential recruitment of autonomic outputs. Indeed, baroreflex function curves reveal a hierarchical activation of sympathetic and parasympathetic limbs, with sympathetic inhibition operating over lower pressure ranges and parasympathetic (cardiac vagal) activation recruited at higher pressures.³⁷ Collectively, these findings support a framework in which carotid and aortic baroreceptors are both mechanistically and functionally differentiated. Aortic baroreceptors exhibit higher mechanosensitivity and preferential engagement of cardiac parasympathetic reflexes, while carotid baroreceptors, although less sensitive, effectively drive sympathetic inhibition

and vascular control. This division of labor, preserved through anatomically and functionally distinct central pathways, enables fine-tuned and flexible regulation of cardiovascular homeostasis under varying physiological and pathophysiological conditions.

Axonal and Terminal Diversity of Baroreceptors: Linking Structure to Firing and Threshold

Baroreceptor nerve endings display a modular organization with interacting functional domains that support mechanosensory control of blood pressure. A mechanotransduction domain converts arterial stretch into receptor potentials via mechanosensitive ion channels, while a surrounding modulatory domain adjusts sensitivity through GPCR (G protein-coupled receptor)- and mediator-dependent signaling. An electrogenic domain, composed of voltage-gated channels, shapes action potential generation and firing patterns. These signals are then integrated and transmitted centrally, forming a frequency-encoded output that drives baroreflex regulation (Figure 2).

Despite their unifying role, baroreceptors are remarkably heterogeneous in axonal type, firing properties, and transduction mechanisms. This diversity enables the baroreflex to encode both phasic and tonic features of arterial pressure. Below, we integrate current understanding of axonal classes, firing classifications, and the emerging evidence on end-net versus flower-spray nerve terminals, which refine the structure-function relationship of baroreceptors.

Axon Class, Threshold, and Firing Archetype

Baroreceptor axons are broadly divided into A-type (myelinated) and C-type (unmyelinated) fibres. A-fibres have large diameters, rapid conduction velocities (10–30 m/s), and typically mediate phasic responses to transient changes in arterial pressure. C-fibres, by contrast, are smaller, slower (≤ 2 m/s) and exhibit tonic or irregular discharge, often firing at baseline pressures even in the absence of overt stretch stimuli^{26,38–40} (Table).

Classic single-fibre recordings revealed 2 archetypal pressure-discharge relationships, termed type I and type II fibres.⁴¹ Type I (A-fiber) baroreceptors exhibit higher sensitivity (dynamic gain), steeper stimulus-response curves, and rapid, phasic increases in firing (Figure 3), making them ideally suited for detecting beat-to-beat or dynamic changes in arterial pressure.⁴¹ In contrast, type II (predominantly C-fiber) baroreceptors show lower sensitivity, wider operating ranges, and tonic baseline discharge (tonicity; Figure 3), indicating a role in encoding absolute or mean arterial pressure rather than rapid fluctuations (Table). Importantly, type I fibres are more susceptible to acute resetting, a rightward shift in their pressure-response curve during sustained hypertension,

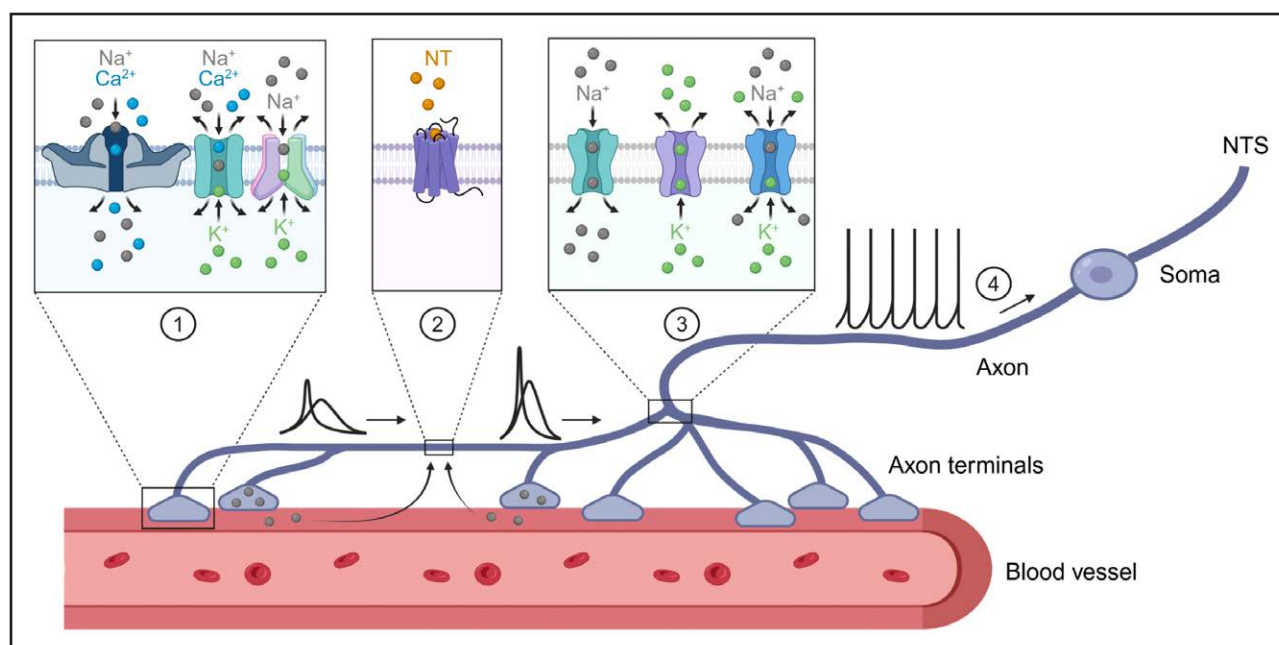


Figure 2. Schematic overview of the functional domains of baroreceptor nerve endings.

Baroreceptor nerve terminals exhibit a modular organization composed of distinct but interacting molecular domains that together support the mechanosensory control of arterial blood pressure. (1) Mechanosensory transduction domain: The peripheral terminal membrane (encoder) contains mechanosensitive ion channels that convert arterial wall stretch into receptor potentials, representing the primary site of mechanical-to-electrical transduction. Each encoder is proposed to generate depolarizing signals that are conveyed through daughter afferent axons for further integration. (2) Modulatory domain: Surrounding the mechanotransduction machinery is a repertoire of GPCRs (G protein-coupled receptors) and hormone/neuropeptide receptors that may dynamically regulate baroreceptor sensitivity. These include receptors for mediators such as angiotensin II, endothelin, bradykinin, serotonin, norepinephrine, and prostaglandins. Efferent release of transmitters by baroreceptor endings can also feed back to modulate neuronal signaling. Through second-messenger pathways, these inputs modulate channel gating, alter terminal excitability, and integrate humoral and autonomic signals. (3) Electrogenesis domain: Voltage-dependent ion channels distributed along the terminal membrane and afferent segments shape action potential initiation and firing patterns. These include voltage-gated sodium (Nav), potassium (Kv), and HCN (hyperpolarization-activated cyclic nucleotide-gated) channels, among others. This domain determines the gain and dynamic range of baroreceptor output. (4) Integration and afferent transmission domain: Electrogenic signals arising from multiple terminal regions are integrated and encoded into a frequency-dependent output transmitted via the glossopharyngeal nerve (Hering nerve) to the nucleus tractus solitarius (NTS). This integrated signal contributes to the regulation of the arterial baroreflex. Created with BioRender.com.

whereas type II fibres are relatively resistant.⁴³ This allows the reflex arc to maintain both sensitivity to transient changes and stability across chronic pressure shifts.

Ionic Mechanisms Underlying the Distinct Firing Properties of A- and C-Type Baroreceptors

Although both A-type (myelinated) and C-type (unmyelinated) baroreceptor subtypes encode arterial pressure, their divergent discharge profiles reflect fundamental biophysical distinctions in excitability.^{44–46} Electrophysiological and computational studies have begun to highlight how differences in ion-channel expression and kinetics underlie this A/C dichotomy (Table).

Sodium Channel Specializations and Spike Initiation

A hallmark of A-type (myelinated) baroreceptors is their ability to initiate spikes at relatively hyperpolarized voltages and to sustain high-frequency excitability. This is

most parsimoniously explained by a high density of fast, tetrodotoxin-sensitive Na⁺ channels (eg, isoforms, such as Nav1.7 or related subtypes).⁴⁷ These channels typically activate rapidly, inactivate quickly, and support steep rising phases of the action potential, thereby minimizing the threshold and allowing rapid depolarization.⁴⁸ In sensory neurons more broadly, Nav1.7 is often described as a threshold channel, contributing to perithreshold inward current, while other fast Na⁺ channels take over in the upstroke.⁴⁹ This channel profile enables low firing thresholds and high-frequency spiking in response to dynamic inputs.⁴⁶ In addition, the presence of large hyperpolarization-activated HCN1 current in A-type neurons tends to increase baseline excitability and allow faster recovery.⁴⁶ This supports the reduced braking in A-type neurons.

In contrast, C-type sensory neurons, including nociceptors and many unmyelinated afferents, frequently express tetrodotoxin-resistant Na⁺ currents (eg, Nav1.8, Nav1.9). Nav1.7 is expressed at a lower density in C-type baroreceptors and in conjunction with tetrodotoxin-insensitive Nav1.8 and Nav1.9.⁴⁶ Because of their slower activation

Table. Comparative Features of A- Versus C-Type Baroreceptor Afferent

Feature	A-type fibres (often type I)	C-type fibres (often type II)
Axon	Myelinated	Unmyelinated
	Large diameter	Small diameter
Conduction velocity	Fast (10–30 m/s)	Slow (<2 m/s)
Threshold	Higher mechanical threshold	Lower mechanical threshold
	Low electrical threshold	Depolarized electrical threshold
Baseline activity	Usually silent until the threshold	Often spontaneous tonic firing
Firing pattern	Abrupt onset, steep slope	Gradual onset
	High max frequency	Broad range
		Lower max frequency
Resetting	Pronounced acute resetting	Relatively resistant to resetting
Na ⁺ currents	Fast, TTX-sensitive (Nav1.7)	Prominent TTX-resistant (Nav1.8)
K ⁺ currents	Lower density	Higher density
	Supports high frequency	Promotes adaptation
Functional role	Rapid buffering of pressure transients	Encoding tonic mean pressure
		Baseline reflex drive

TTX indicates tetrodotoxin.

and voltage dependence, tetrodotoxin-resistant currents impose a higher effective threshold ($\approx 15\text{--}20$ mV more depolarized in the case of Nav1.8) and slower upstroke velocities (often 5–10 $\times$ slower) relative to fast Na⁺ currents. These features reduce the maximal instantaneous firing rate and blunt the sharpness of dynamic encoding.⁴⁶ In sum, C-type neurons accept a lower-gain, slower response regime in exchange for robustness and stability under sustained depolarization.

It is worth noting that neuronal excitability is often robust to manipulations of a single Na⁺ subtype owing to compensatory contributions from other subtypes. In nociceptors, for example, blocking Nav1.7 may have a limited effect if Nav1.8 or Nav1.3 is upregulated (or vice versa).^{50,51} Such degeneracy implies that in baroreceptor neurons, multiple Na channel isoforms might contribute partially, and that shifts in relative expression may tune excitability. The core principle, however, remains: fast, steep Na⁺ currents favor low threshold and high frequency, contrasted against slower Na⁺ currents that enforce a raised threshold and slower kinetics. Together, the dominance of fast versus slow Na⁺ channel subtypes provides a fundamental axis of divergence: A-fibers maximize temporal precision and phasic responsiveness, while C-fibers prioritize tonic encoding capacity.

Potassium Currents, Braking, and Firing Adaptation

In C-type baroreceptor neurons, the presence of non-inactivating (delayed-rectifier) K(v) channels, Ca²⁺-activated K⁺ (K_{Ca}) channels, and possibly Kv7 channels is hypothesized to be relatively prominent.^{45,46,52} These outward conductances play a braking role: they slow the return of the membrane potential toward baseline,

deepen or prolong afterhyperpolarizations, and introduce spike-frequency adaptation or accommodation during sustained input.⁴⁶ In doing so, they constrain firing frequency, mitigate excitotoxicity or runaway firing, and favor tonic, steadier discharge over phasic bursts. In contrast, A-type baroreceptors are thought to express lower densities of stabilizing K⁺ conductances or favor more rapidly inactivating K⁺ currents (eg, A-type transient K currents) with limited prolonged influence.⁴⁶ By reducing the braking effect, A-fibers can repolarize quickly, recover sodium channels more rapidly from inactivation, and generate successive action potentials with minimal accommodation, thereby preserving high-gain dynamic responses (Table).

The power of computational modeling is that it allows systematic exploration of how combinations of Na⁺ and K⁺ conductances yield distinct A- or C-type firing regimes.⁵³ The seminal work by Sturdy et al⁴² built a biophysically grounded model in which arterial wall mechanics, nerve-ending strain (mechanotransduction), and Hodgkin–Huxley-style ionic dynamics are coupled. Through parameter tuning, their simulations matched empirical firing curves of both A- and C-type baroreceptors (eg, step and ramp pressure stimuli). In that model, A-type firing was reproduced by higher fast Na⁺ conductance and lower K⁺ conductance relative to C-type. C-type behavior required reduced fast Na⁺ conductance, a shift toward slower Na currents, and increased K⁺ conductance to impose adaptation and reduce maximal firing. Sensitivity analyses further showed that altering K⁺ conductance slopes or densities moved the model between phenotypes in a smooth conductance space, consistent with the idea of a continuum of firing phenotypes.⁴² Thus, the relative weighting of Na⁺ depolarizing drive versus K⁺ stabilizing currents defines excitability landscapes: A-type neurons

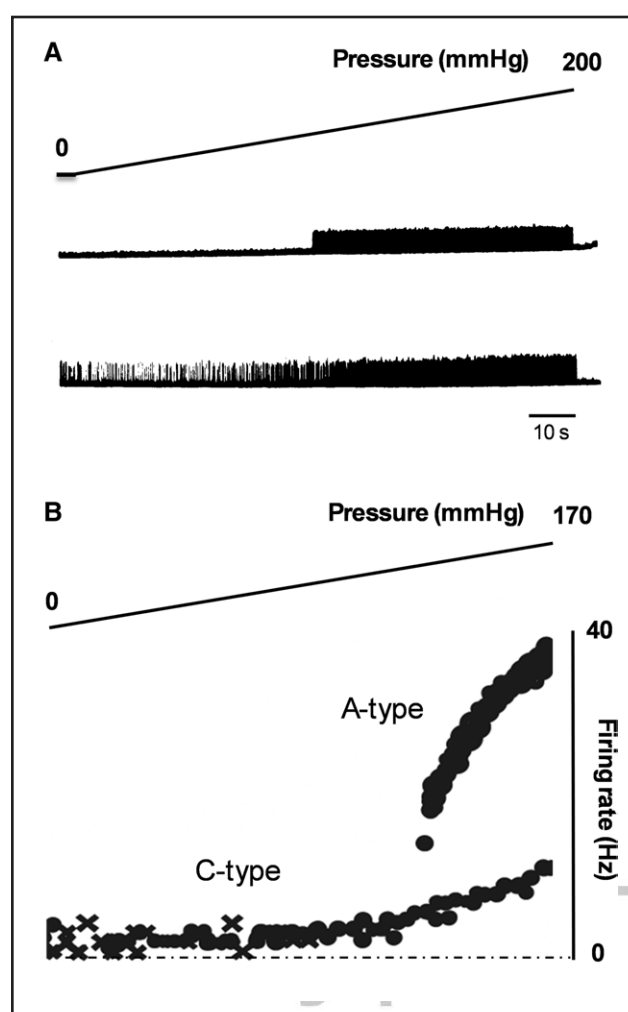


Figure 3. Distinct activation properties of A-type and C-type baroreceptors.

A, Comparison of the characteristic firing properties of A-type and C-type baroreceptor afferents in response to arterial pressure changes in the dog carotid sinus. The figure illustrates the differences in activation threshold, firing frequency, and dynamic range that distinguish the 2 principal barosensory fiber classes. Data from Seagard et al²⁶ with permission. Copyright © 1990, Wolters Kluwer Health Inc. **B**, A mechanical ramp stimulus (**bottom**) and the associated firing rate response for modeled A- and C-type baroreceptors from a carotid artery. Adapted from Sturdy et al⁴² with permission. Copyright © 2017, Springer Nature.

tilt toward high excitability, while C-type neurons enforce stability and adaptation. These models reinforce the view that ion-channel complement and conductance balance can, on their own, explain much of the divergence of A-versus C-type firing, even before considering differences in mechanotransduction coupling.

Terminal Morphologies: End-Net Versus Flower-Spray Endings

Classical anatomic studies have identified distinct baroreceptor terminal morphologies, including end-net and flower-spray endings, in the aortic arch. More recent

genetic and anatomic mapping has demonstrated that this morphological diversity is also present in the carotid sinus, indicating that these terminal types are not restricted to a single barosensory territory^{23,54} (Figure 4). These structures have been described across multiple species (mouse, rat, dog), although often without clear functional discrimination.

End-net terminals form fine, branching networks that extend along the vessel circumference. Functionally, they exhibit low mechanical thresholds, high sensitivity, and steep gain, consistent with type I/A-like baroreceptor profiles. Their geometry and distribution make them well-suited to detect small, rapid changes in wall strain and to support phasic reflex buffering (Table S1). Ablation of these neurons abolishes baroreflex responses, establishing end-net endings as a core substrate for baroreceptor transduction.^{13,23,55}

In contrast, flower-spray endings are typically described as compact, bush-like terminals embedded within the adventitial connective tissue, in close association with collagen, elastin, and supporting cells.²⁴ They are particularly abundant in the carotid sinus and atrial regions. Thus, although end-net and flower-spray terminals coexist within the same vascular territories²¹ (aortic arch and carotid sinus), often within the adventitial layer, they differ markedly in morphology and function. Recent work in mice further indicates that flower-spray endings are not essential for basic baroreception (see below), suggesting they may represent higher-threshold or auxiliary sensors engaged during large or sustained stretch, or subserving distinct sensory modalities.

Electrophysiological recordings from single baroreceptor afferents have long reported multi-phasic recruitment curves. Instead of a smooth, monotonic increase in firing rate with pressure, some fibres display 2 distinct thresholds or inflection points.^{26,56,57} These inflections correspond well with the idea that low-threshold end-net elements are recruited first, followed by higher-threshold ones as distension increases. In addition, the firing dynamics differ within the same axon: rapidly adapting, high-gain responses are prominent at low pressures, while more sustained or irregular discharges emerge at higher pressures.⁵⁸

These findings support the multiple sensor model, in which a single baroreceptor fiber integrates input from morphologically and functionally distinct mechanosensory specializations^{27,28} (Figure 4). Rather than functioning as a uniform sensor, the afferent operates as a composite unit (1) with a low threshold, fast activation, likely responsible for initial sensitivity around the operating point of the baroreflex (end-nets) and (2) a higher threshold, sometimes slower or adaptive, recruited during stronger arterial distension (flower-sprays?). The net result is a firing pattern with multiple recruitment points, matching observed composite firing curves. This arrangement provides important advantages. First, it extends the

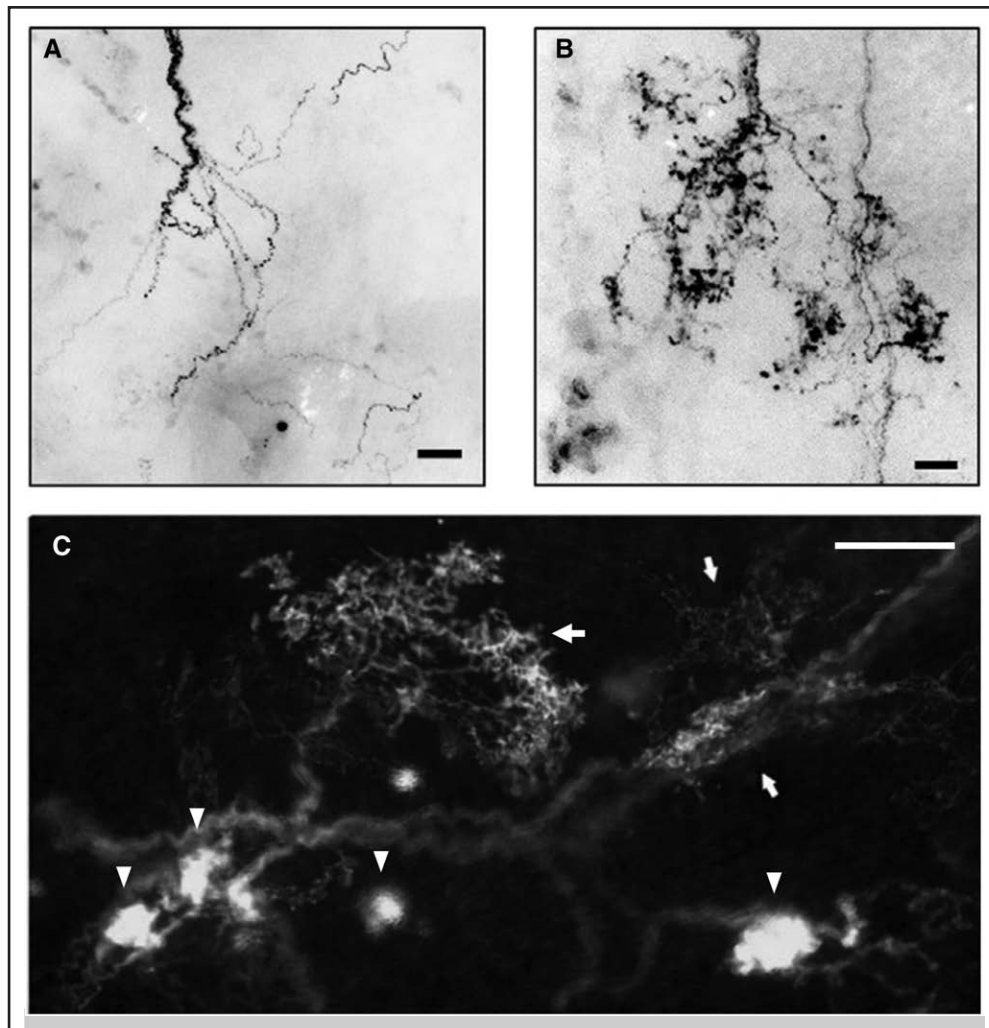


Figure 4. Morphology of baroreceptor nerve endings in the aortic arch.

A and **B**, Representative examples of end-net endings from PIEZO2-expressing NJP neurons (**A**, Piezo2-ires-Cre x AAV-flex-tdTomato) and flower-spray terminals from MC4R-expressing NJP neurons (**B**, Mc4r-2a-Cre x AAV-flex-tdTomato) in mouse aortic arches. Scale bars, 30 μ m. Adapted from Min et al.²³ **C**, Two types of flower-spray endings are connected to myelinated afferents of the rabbit aortic arch: solid, compacted, and ball-like structures (arrowheads) and much larger, extended diffuse-type structures (arrows). They differ morphologically as shown by the double Na⁺/K⁺-ATPase and myelin binding protein staining. Scale bar, 100 μ m. Data adapted from Liu et al.²⁷

dynamic range of individual fibres, enabling encoding of both diastolic-level changes and systolic excursions without saturating at either end. Second, it reduces reliance on averaging across large populations of fibres to achieve graded sensitivity—instead, each axon carries a built-in 2-step (or multistep) sensor system. Finally, the coexistence of distinct mechanosensory endings on 1 axon may permit differential modulation (eg, by local stretch distribution, hormones and neuromodulators [angiotensin II, estrogen] or pathology), further diversifying baroreflex input.

Although the multiple-sensor model reconciles anatomic, molecular, and electrophysiological observations that cannot be explained by a 1-ending 1-fiber framework, there is as yet no definitive electrophysiological evidence that a single baroreceptor axon intrinsically possesses 2 distinct mechanosensory threshold mechanisms in the

classical sense of independent channel mechanisms at its terminal sites.

PIEZO2 and the Baroreflex: End-Nets in Focus, Flower-Sprays in Question

The strongest, direct evidence that PIEZO channels sit at the root of arterial mechanotransduction comes from genetic loss- and gain-of-function studies in mice (Table S2). Zeng et al¹³ used conditional deletion and optogenetic activation of Piezo-expressing vagal/glossopharyngeal afferents to show that combined loss of Piezo1+Piezo2 abolishes aortic depressor nerve activity and baroreflex responses, while selective activation of Piezo2-positive afferents can drive the reflex.¹³ These experiments make a compelling case that PIEZOs are necessary and sufficient, at least in the models tested,

for acute baroreceptor signaling. Independent anatomic and genetic work reinforced that conclusion by mapping a distinctive Piezo2-marked population of nodose/petrosal neurons with aortic claw endings that surround the aortic arch and exude fine end-net endings.²³ Ablation of these Piezo2-expressing neurons markedly impairs baroreflexes. Other arterial sensory neurons that form flower-spray terminals appear dispensable for baroreception. Together with the loss-of-function data, these morphological and functional observations argue that Piezo2 specifically labels bona fide baroreceptor afferents.²³

However, the story is not settled. Several groups have flagged important gaps and alternative interpretations.^{59–61} As discussed by Stocker et al,⁵⁹ several methodological and conceptual aspects of the study by Zeng et al,¹³ warrant further clarification before concluding that PIEZO1/2 are the sole mechanotransducers of baroreception. These include the need to better define the localization of Piezo1/2 at sensory terminals and to clearly distinguish baroreceptors from other vagal mechanosensitive afferents. In addition, the interpretation of blood pressure measurements in knockout or pharmacologically treated models, the lack of direct aortic depressor nerve stimulation to assess central processing, and the limited evaluation of blood pressure variability leave some aspects of baroreflex function open to further investigation. In short, the genetic loss-of-function data are compelling but do not yet close every mechanistic loophole.^{59–61} (Table S2).

However, strong evidence is accumulating pointing to Piezo2 as the predominant mechanosensory channel driving baroreceptor function, whereas Piezo1 appears to play only a minor role. In models of hypertension, including spontaneously hypertensive rats and Ang II- or L-NNA-induced disease, Piezo2 expression is selectively diminished in both ganglionic neurons and aortic baroreceptor endings, a deficit not mirrored by Piezo1.⁶² Loss of Piezo2 function, whether through genetic knockdown or pharmacological inhibition, leads to impaired baroreflex sensitivity, elevated arterial pressure, and increased sympathetic outflow, firmly establishing its causal role in blood pressure homeostasis. Mechanistic insights further reveal that Nedd4-2 (neural precursor cell expressed developmentally downregulated 4-like) directly regulates Piezo2 via ubiquitin-mediated degradation, and its upregulation in hypertensive states contributes to Piezo2 downregulation, linking altered mechanotransduction with the pathogenesis of hypertension. Other recent functional studies complicate the simple dichotomy PIEZO2 = barosensor, PIEZO1 irrelevant. Work from multiple groups reports that Piezo1 can modulate baroreflex transmission, for example, Piezo1 activation facilitates afferent neurotransmission in nodose/ nucleus tractus solitarius pathways and may have sexually dimorphic and condition-dependent roles that interact with Piezo2 expression.⁵⁵ These data support a model

in which Piezo2 marks primary baroreceptor afferents while Piezo1 can contribute at multiple levels (afferent endings, synaptic transmission, or vascular/endothelial interactions), and in some contexts the 2 channels act redundantly or synergistically.⁵⁵

Thus, the parsimonious, current synthesis is that PIEZO2 identifies a principal population of arterial mechanosensory afferents required for the canonical baroreflex in mice, while PIEZO1 plays a modulatory or complementary role that may vary by cell type, vascular bed, sex, and pathophysiological state.

There is direct human tissue evidence (immunohistochemistry, proteomics) that PIEZO1 and PIEZO2 proteins are present in the carotid sinus/aortic adventitial nerves.^{30,63} Thus, PIEZOs are a strong candidate baroreceptor mechanotransducer in humans, and mutations could be risk factors, but confirmed human cases linking PIEZO mutations to baroreflex dysfunction have not yet been reported. Multiple case series and reviews describe human PIEZO2 loss-of-function and gain-of-function phenotypes, for example, congenital proprioception/touch loss, arthrogryposis, respiratory problems, but these reports do not present robust baroreflex testing or definitive BP-baroreflex phenotypes tied to PIEZO mutations.⁶⁴ Critical open questions remain: (1) do mechanically activated PIEZO currents exist at native baroreceptor terminals under physiological stretch magnitudes; (2) what is the precise contribution of endothelial PIEZO1 (shear/flow sensing) versus neuronal PIEZO1 to reflex gain; (3) how conserved are these roles in larger mammals and humans; and (4) can residual baroreflexes after single-gene deletions be explained by true redundancy, developmental compensation, or incomplete targeting?

The Hunt for Alternative Baroreceptor Mechanosensors

PIEZO2 proteins are the leading candidate for acute pressure sensing because they are required for the arterial baroreflex, and their expression is localized to end-net but not flower-spray endings.²³ Nevertheless, the mechanosensory role of flower-spray endings remains unresolved. A plausible hypothesis is that flower-spray endings may employ 1 or more of these alternative channels to detect hemodynamic forces. Their morphological resemblance to other mechanoreceptor types in ultrastructural studies strengthens the possibility that they contribute to baroreception through a PIEZO2-independent mechanism.

TRPC5: a Modulator of Baroreflex

The TRPC5 channel is expressed in peripheral and central neurons, including sensory neurons that innervate the aortic arch and carotid sinus. TRPC5 can be opened by hypo-osmotic stress and by membrane stretch, producing

robust Ca^{2+} entry; this mechano-sensitivity depends on intact actin cytoskeleton coupling and, at least in many experimental preparations, occurs independently of classical PLC (phospholipase C) signaling.^{65,66}

It has been proposed as a mechanotransducer that contributes to transducing arterial stretch into afferent firing.¹⁵ Immunoreactive TRPC5 signals were detected in rat nodose neurons and in putative baroreceptor nerve terminals innervating the adventitia of the aortic arch.¹⁵ Trpc5 knockout and dominant-negative TRPC5 expression reduced pressure-induced afferent firing and blunted baroreflex-mediated heart-rate responses; telemetry showed increased short-term BP lability in knockout mice, supporting a physiological role in stabilizing rapid BP fluctuations.¹⁵

Although these data suggest a role for TRPC5 in acute baroreflex function, several authors have raised technical concerns and submitted critical correspondence questioning aspects of the electrophysiological analyses and single-channel interpretations in the original work.⁶⁷ Reanalysis and subsequent discussion⁶⁸ indicate that some single-channel observations may have been influenced by experimental enhancers, and that the precise molecular composition of the native channel(s) remains uncertain. In addition, these follow-up data report no difference in baseline blood pressure between WT and TRPC5 knockout mice. Taken together, these findings underscore the need for further investigation before concluding that TRPC5 plays a definitive role in cardiovascular regulation or that it is directly activated by physiological mechanical forces in baroreceptors, particularly relative to other candidate mechanisms.⁶⁷ Future studies employing conditional (adult-onset) and cell type-specific Trpc5 deletions, thereby minimizing developmental compensation and distinguishing effects at afferent terminals versus cell bodies or central circuits, will likely be necessary to resolve these issues (Table S3).

Recent findings also raise the possibility that TRPC5 operates not as a stand-alone mechanosensor but within heteromeric channel complexes, particularly with TRPC1.⁶⁹ TRPC1, once considered an independent channel, may instead act as a regulatory subunit modulating TRPC5's gating and mechanical sensitivity. This hypothesis aligns with emerging structural and electrophysiological data, suggesting a TRPC5/TRPC1 heteromeric assembly could constitute a more physiologically relevant mechanosensor than TRPC5 alone.

It is worth considering that the reported effect whereby GPCR signaling potentiates the osmo-mechanical activation of TRPC5 channels may represent a modulatory rather than a primary transduction mechanism.^{70,71} In this context, activation of GPCR pathways (particularly those coupled to Gq/11 and leading to diacylglycerol or other second messenger generation) could enhance the sensitivity of TRPC5 to membrane stretch or hypo-osmotic stress, without TRPC5 itself serving as the direct

mechanotransducer. Thus, TRPC5 may function more as an amplifier or integrator of mechano-chemical signals within baroreceptor afferents, rather than as the principal sensor of arterial pressure changes. This interpretation aligns with findings showing that TRPC5 deletion increases short-term blood pressure variability without affecting mean arterial pressure, supporting a modulatory role in short-term baroreflex function rather than primary mechanical transduction.

Many loci are associated with blood-pressure traits in humans, but TRPC5 has not been identified as a clear human BP/ baroreflex locus in the major BP GWAS catalogues.^{72,73} Although this is an absence of evidence rather than proof of absence, the data suggest that TRPC5 channels (or TRPC5 mutations/variants) are not major contributors to baroreceptor function or baroreflex performance in humans.

TRPV1: a Controversial Candidate in Peripheral Baroreflex Control

TRPV1 (transient receptor potential vanilloid 1) is a nonselective cation channel traditionally recognized for its role in nociception and pain modulation.⁷⁴ Classically, TRPV1 is activated by noxious heat ($>43^{\circ}\text{C}$), protons, and endogenous inflammatory mediators such as bradykinin, anandamide, and lipoxygenase metabolites released during tissue injury or ischemia.^{75,76}

TRPV1 immunoreactivity has been found in NF200-positive fibers, possibly corresponding to baroreceptor endings surrounding the aortic arch.¹⁶ Functional support for the role of TRPV1 in peripheral baroreflex or baroreceptor-related reflexes comes primarily from rodent work using pharmacology, targeted ablation, and genetic knockout. Ablation of TRPV1-expressing afferents with resiniferatoxin or pharmacological TRPV1 blockade has been reported to impair baroreceptor-mediated reflex responses^{76,77} while activation facilitates carotid sinus baroreceptor activity.⁷⁸ These studies, therefore, implicate TRPV1-positive sensory endings in some aspects of arterial pressure sensing or in the modulation of baroreflex output.

Genetic deletion or long-term ablation experiments give a more complex picture. TRPV1 knockout or chronic ablation in mice has been reported to alter nocturnal (but not diurnal) blood-pressure patterns and to reduce baroreflex sensitivity under certain metabolic challenges,¹⁷ suggesting that TRPV1 may contribute to time-of-day or diet-dependent modulation of autonomic set-points. At the same time, other groups have failed to replicate TRPV1-dependent effects on some pressure reflexes,⁷⁹ highlighting variability between preparations, species, and experimental protocols. In addition, although TRPV1 is robustly expressed in vagal C-fiber neurons,^{80–82} reporter-mouse and physiological studies of primary afferents demonstrate TRPV1 expression/activity in a subset of myelinated A cells.^{83,84} This is relevant when

interpreting vagal data. Differences in fiber type (unmyelinated C versus thinly myelinated A δ), innervation targets, and central termination patterns make it likely that TRPV1 modulation has fiber-specific consequences and therefore that systemic knockouts or broad pharmacology will produce mixed or context-dependent readouts. Recent single-cell and anatomic studies of vagal sensory populations underscore this heterogeneity.^{82,85}

How well this preclinical literature translates to humans is uncertain. Early clinical development of systemically acting TRPV1 antagonists (for analgesia) revealed on-target thermoregulatory side effects (notably with AMG-517) and mixed pharmacodynamic readouts.⁸⁶ The human studies that examined basic cardiovascular variables did not report consistent, clinically meaningful changes in resting or ambulatory blood pressure, although they exposed safety limits for systemic TRPV1 blockade.⁸⁷ Thus, available early-phase human pharmacology does not provide strong support that global TRPV1 blockade substantially alters basal baroreflex control in healthy volunteers. Genetic evidence also fails to provide a clear answer. The common TRPV1 missense variant I585V (rs8065080) has been functionally characterized in vitro and alters channel activity in experimental systems, but large genome-wide association studies for blood-pressure traits have not identified TRPV1 as a replicated, genome-wide significant locus.⁸⁸ Smaller candidate-gene or cohort studies have reported context-dependent or trait-specific associations (eg, with chemosensory phenotypes, salt-taste perception, or modest trait changes in subgroups),^{89–91} but these findings are not consistent across large, well-powered GWAS meta-analyses.⁹² In short, human genetic data do not support a major, population-wide effect of TRPV1 sequence variation on blood pressure at present. These discoveries do argue that TRPV1 is unlikely to be a primary transducer of arterial stretch but that TRPV1 may be a candidate for coupling inflammatory/metabolic cues to autonomic reflexes, modulating baroreceptor function under specific physiological or pathophysiological states (eg, inflammation, renal injury, diet-induced metabolic stress, circadian phases; [Table S3](#)).

ENaC Channels: Potential Mechanosensors With Unresolved Baroreflex Roles

ENaCs, the mammalian members of the degenerin/ENaC (DEG/ENaC) superfamily first defined in *Caenorhabditis elegans*, have long been proposed as mechanosensors because their invertebrate homologs mediate touch and proprioception.^{93,94} Mammalian ENaCs are multi-subunit channels composed principally of α , β , and γ (and, in some tissues, δ) subunits; although α ENaC carries constitutive Na⁺ current in canonical epithelial assemblies, functional heteromers formed with β and γ (and in certain contexts δ) subunits indicate compositional flexibility that may affect mechanosensitivity and gating.⁹⁵

Pharmacology provides some of the most direct functional evidence linking ENaCs to the baroreflex. In a classic set of experiments using isolated carotid sinus preparation, the ENaC/DEG blocker benzamil produced a dose-dependent attenuation of baroreceptor nerve activity and blunted baroreflex function in rabbits.^{18,96} Systemic and local application of amiloride-family blockers similarly reduce baroreceptor signaling and baroreflex gain in multiple species. These pharmacological effects indicate that an amiloride-sensitive Na⁺ conductance participates in arterial mechanotransduction.⁹⁷

At the cellular level, mechanostimulation of nodose ganglion neurons evokes depolarization and calcium entry that are abolished or strongly suppressed by amiloride, pointing to an amiloride-sensitive cation entry pathway in the afferent limb of the reflex.^{19,98} Complementary transcriptional and immunohistochemical studies report $\alpha/\beta/\gamma$ ENaC expression in nodose neurons and in aortic arch baroreceptor terminals, supporting the anatomic substrate for a direct role.^{18,99}

In chronic heart failure (CHF), circulating aldosterone and vasopressin levels are markedly elevated and correlate with reduced baroreflex sensitivity and greater blood-pressure variability.¹⁰⁰ Animal models of CHF demonstrate that ENaC downregulation parallels impaired baroreflex function. In rats with pacing-induced CHF, β - and γ -ENaC protein levels are significantly reduced in nodose ganglia and aortic baroreceptor terminals, accompanied by decreased baroreceptor discharge frequency and attenuated reflex heart-rate responses to changes in arterial pressure.⁹⁹ Comparable reductions in amiloride-sensitive sodium currents have been observed in isolated baroreceptor neurons from CHF animals,¹⁸ supporting a mechanistic link between ENaC expression and afferent excitability. Together, these findings indicate that aldosterone-driven ENaC downregulation may directly compromise baroreceptor signaling, thereby contributing to autonomic imbalance and end-organ injury in CHF. However, the current evidence remains largely correlative. Pharmacological tools such as amiloride and benzamil implicate ENaC in mechanosensory signaling but lack perfect specificity.¹⁰¹ In addition, vascular ENaC contributes to myogenic tone and could modulate baroreflex gain indirectly by altering vessel mechanics.¹⁰² Definitive causality will require cell-specific genetic ablation or rescue of ENaC subunits in baroreceptor afferents, and high-resolution structural studies to clarify whether ENaCs act as a primary mechanically gated pore or as auxiliary regulators within a broader mechanosensory complex that may also include other mechanosensitive channels ([Table S3](#)).

In humans, 3 complementary lines of clinical evidence support a pathophysiological link between ENaC-relevant biology and baroreflex impairment. First, hormonal perturbation and disease cohorts show robust associations between mineralocorticoid excess and reduced baroreflex sensitivity.

Acute experimental elevation of aldosterone in healthy volunteers blunts cardiovascular baroreflex sensitivity, as shown in randomized crossover studies: intravenous aldosterone infusion reduced baroreflex gain measured by the phenylephrine method.¹⁰³ Similarly, Monahan and Leuenberger¹⁰⁴ demonstrated that aldosterone acutely impairs both cardiovascular and sympathetic baroreflex function in humans, consistent with earlier observations showing that aldosterone attenuates the baroreflex response in humans.¹⁰⁵

Patients with biochemical mineralocorticoid excess (primary aldosteronism) exhibit lower baroreflex sensitivity and greater short-term blood-pressure variability than matched essential hypertensive controls.¹⁰⁶ Treatment with mineralocorticoid receptor antagonists or successful adrenalectomy partially restores autonomic indices in these patients.¹⁰⁷ This human data establishes a reproducible endocrine-reflex association implicating aldosterone-sensitive pathways for which ENaC is the canonical downstream effector in epithelia. Second, clinical pharmacology and therapeutics provide supporting, but nondefinitive, evidence. Systemic ENaC inhibitors (amiloride-class diuretics) predictably alter renal sodium handling, plasma volume, and long-term blood pressure, but human studies isolating the acute, local effects of ENaC blockade on baroreflex gain are rare. Thus, pharmacology supports biological plausibility but does not demonstrate that ENaC in baroreceptor terminals functions as a local mechanosensor. Third, human genetic and observational studies are supportive but incomplete. Germline ENaC gain- and loss-of-function mutations (eg, Liddle syndrome and pseudohypoaldosteronism type I) strongly affect systemic blood pressure and extracellular volume, underscoring the physiological importance of ENaC in cardiovascular homeostasis.¹⁰⁸ However, systematic phenotyping of baroreflex sensitivity in well-characterized ENaC mutation carriers, controlling for volume status and antihypertensive therapy, remains sparse.

Taken together, the human evidence establishes a clinically important association: mineralocorticoid excess and ENaC-related physiology track with depressed baroreflex function and worse cardiovascular outcomes. However, the evidence is predominantly correlative. Direct mechanistic proof in humans, that is, demonstration of ENaC protein localized to human baroreceptor terminals together with acute, local amiloride-sensitive changes in single-unit afferent firing or reflex gain, has not yet been reported in the literature accessible to date.

ASIC Channels as Acid-Tuned Modulators of Cardiovascular Mechanotransduction

ASICs (ASIC1–4) belong to the ENaC/degenerin superfamily and constitute Na⁺-permeable channels widely expressed across neuronal and neuroepithelial tissues. Although classically recognized as extracellular

pH sensors, recent work implicates ASICs in mechanosensory circuits relevant to baroreflex and chemoreflex control. Nevertheless, accumulating evidence suggests that ASICs, especially ASIC2, do not act as stand-alone mechanosensors. Instead, they appear to function as acid-tuned modulators embedded in multiprotein signaling architectures that integrate mechanical and chemical cues to regulate cardiovascular homeostasis (Figure 5).

Within the nodose ganglion, transcripts encoding ASIC1a, 1b, 2a, 2b, and ASIC3 are abundant, with ASIC2b typically dominating expression levels.^{20,109} ASIC1, 2, and 3 were present and colocalized with neurofilament-L antibody in baroreceptor nerve terminals in mice.²⁰ Genetic deletion of ASIC2 elicits a prominent cardiovascular phenotype characterized by reduced baroreflex sensitivity, diminished mechanosensitivity of nodose afferents, elevated sympathetic tone, and hypertension.^{20,110} These functional deficits suggest that ASIC2 supports mechanotransduction in nodose afferents.

However, the role of ASIC2 is not without contention. In dorsal root ganglion neurons, deletion of ASIC2 and ASIC3 often fails to alter mechanically-evoked currents, suggesting that those ASICs are not essential for direct force sensing in all sensory contexts.¹¹¹ Moreover, ASIC2^{-/-} mice preserve the initial burst of baroreceptor afferent firing in response to rising blood pressure, suggesting that ASIC2 is more critical for sustained or adaptive mechanotransduction than for the immediate onset response.¹¹² These nuanced discrepancies favor a model in which ASIC2 must cooperate with accessory scaffolding proteins or parallel force-gated pathways to manifest mechanosensitivity *in vivo*. Importantly, ASIC2 colocalizes with β ENaC and γ ENaC subunits in cerebral arteries and vascular smooth muscle cells, and ASIC2 deletion abolishes pressure-induced constriction in artery preparations, implying that ASIC2 participates in vascular mechanotransduction complexes.¹¹³ Indeed, ASIC2 (often in heteromeric assemblies) is postulated to modulate mechanosensitivity in vascular tissues by interacting with ENaC subunits and cytoskeletal/connective matrix components.¹¹³

Beyond baroreceptor circuits, ASIC2 exerts influence over chemoreflex pathways. ASIC2^{-/-} mice display enhanced chemoreflex sensitivity, and in spontaneously hypertensive rat models, nodose ganglionic ASIC2 expression is reduced in conjunction with amplified chemoreceptor drive.¹¹⁰ This suggests ASIC2 may act as a cross-regulatory node, damping sympathetic overactivation by coordinating baroreceptor and chemoreceptor interplay in disease states marked by sympathetic dominance and baroreflex impairment.

In sum, ASIC2 is not a traditional force-gated channel; rather, it behaves as a multimodal signal integrator, converting local acid microenvironments into modulatory control of mechanosensory gain within cardiovascular reflex circuits (Figure 5). This model reconciles

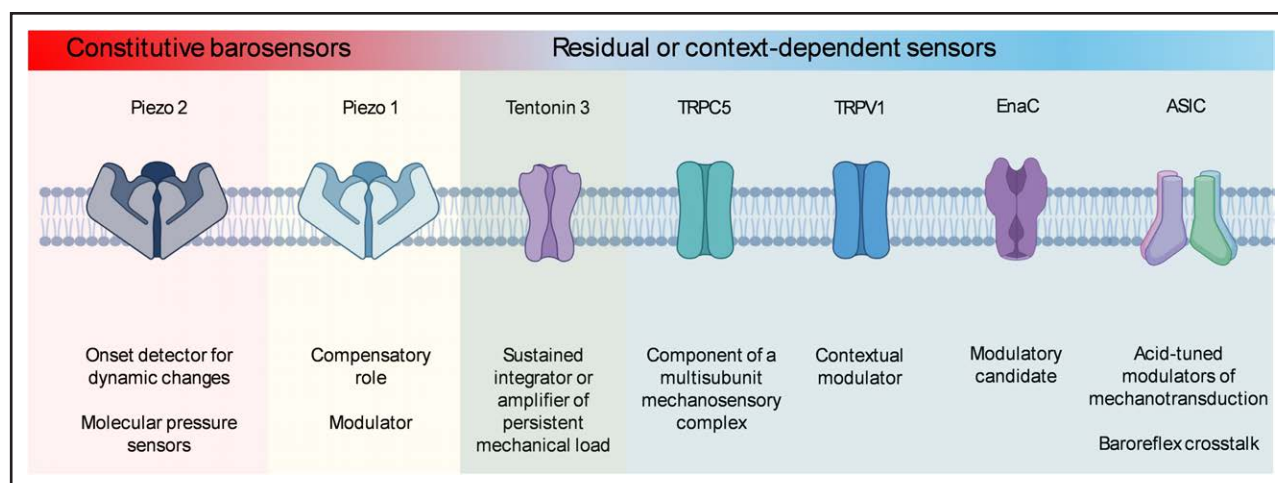


Figure 5. Conceptual hierarchies of mechanosensitivity in arterial baroreception.

The figure depicts candidate baroreceptor ion channels along a mechanotransduction confidence gradient, represented by a warm-to-cool color scale. Red-toned elements highlight channels with strong, direct evidence for bona fide mechanotransduction, typified by Piezo channels, which meet the criteria for force-from-lipid gating and are essential for baroreflex initiation. Piezo2 is proposed as a leading candidate for fast onset detection of dynamic mechanical changes, whereas Piezo1 may contribute at multiple subcellular levels (afferent endings and synaptic transmission) and is likely to play a modulatory or compensatory role. Blue-toned elements represent less certain candidates, including channels, such as tentonin-3, TRPC5 (transient receptor potential canonical 5), TRPV1 (transient receptor potential vanilloid 1), ENaC (epithelial sodium channel)/DEG (degenerin) subunits, and ASICs (acid-sensing ion channel), which exhibit stretch- or context-dependent activation but lack definitive evidence for primary mechanical gating. Tentonin3, with its slow kinetics, appears better suited to sustained encoding than to rapid phasic responses, suggesting a role as an amplifier of persistent mechanical load. Other blue candidate channels are depicted as modulatory or auxiliary contributors, influencing excitability, sensitization, or mechanotransduction primarily under specific physiological or pathological conditions. Created with BioRender.com.

conflicting *in vivo*, *ex vivo*, and *in vitro* data and reframes ASIC2's role: acid-sensitive amplifiers of mechanical input, rather than independent mechanosensors. Dissecting the molecular interactome of ASIC2, mapping its cell-type-specific partners, and resolving its temporal role in mechanotransduction will be key to understanding how ASICs contribute to cardiovascular homeostasis and disease.

TMEM150C (Tentonin3/TTN3) Is a Promising Candidate, But Gaps Remain

Arterial baroreceptors must transduce slowly varying vessel stretch into maintained afferent firing over the cardiac cycle, favoring mechanosensors with slow inactivation kinetics and sustained responses. Among candidate proteins, TTN3 has emerged as an intriguing contender. In 2020, Lu et al¹⁴ proposed that TTN3 contributes to sensing arterial pressure in the aortic arch and nodose afferents, based on *in vivo* cardiovascular phenotypes in knockout mice and terminal expression mapping. Subsequent functional and structural work showing pore-forming behavior has reinvigorated interest in TTN3 as a slow mechanosensitive channel¹¹⁴ (Figure 5).

Supporting a baroreflex role, Lu et al¹⁴ showed that TTN3 is expressed in nerve terminals resembling end-nets within the aortic arch adventitia and in nodose ganglion neurons, and that TTN3 deletion yields hypertensive

phenotypes, destabilized arterial pressure, tachycardia, and reduced baroreflex sensitivity. They further demonstrated that chemogenetic silencing or activation of TTN3⁺ nodose neurons altered blood pressure and heart rate in the direction predicted for baroreceptor effectors, and that overexpression of TTN3 in the nodose of knockout mice rescues the phenotype. These data, while correlative, provide strong *in vivo* support that TTN3 contributes to baroreflex control. Earlier, Hong et al^{115,116} had reported that heterologous expression of TTN3 in cell lines confers mechanically activated currents with slow inactivation kinetics (distinct from Piezo2) and proposed TTN3 as a slowly adapting mechanosensor in sensory neurons. They also presented mutational and coexpression experiments to argue for pore formation (or at least modulation). A pivotal 2024 report by Pak et al¹¹⁴ purified TTN3 protein, reconstituted it into lipid bilayers, and observed spontaneous and pressure-sensitive channel currents with slow inactivation kinetics. They inferred a tetrameric architecture and identified constriction sites whose mutations altered the mechanosensitive currents. This is perhaps the strongest evidence to date that TTN3 can act as a pore-forming mechanosensitive subunit.

A significant critique arose from a Matters Arising by Dubin et al¹¹⁷ who reported that the mechanically induced currents attributed to TTN3 in heterologous systems disappeared when endogenous PIEZO1 was genetically ablated (CRISPR-Cas9 knockout). In that light, they argue that TTN3 might act as a modulator

of PIEZO1 rather than an independent mechanotransducer. This raises the pervasive concern that endogenous mechanosensitive currents in expression systems (eg, residual PIEZO channels) can confound interpretation. Some independent laboratories have failed to replicate a requirement for TTN3 in mechanosensitivity in native sensory neurons (eg, cutaneous afferents), casting doubt on the universality of its mechanotransducer function. A paper by Ojeda-Alonso et al¹¹⁸ explicitly questions participation of TTN3 in nodose neuron mechanosensitivity. The original *in vivo* work of Lu and colleagues¹⁴ has not yet been widely independently replicated, particularly in baroreceptor nerve firing. Independent corroboration is therefore needed, and it is not yet clear whether TTN3's functional role is primary force detection, mechanical gain amplification, or regulation of excitability in baroreceptor afferents.

Currently, there is no direct functional evidence implicating TMEM150C in human baroreceptor activity. Immunohistochemical analysis has detected TTN3 immunoreactivity in subsets of human dorsal root ganglion neurons and in peripheral mechanosensory endings, including Meissner, Pacinian, and Ruffini corpuscles as well as muscle spindles.¹¹⁹ These findings indicate that TTN3 is expressed in some human sensory neurons, consistent with a potential role in mechanosensation. However, the study did not localize TTN3 to aortic or carotid baroreceptor terminals, nor did it provide functional evidence of a baroreceptive role in humans. Complementary multi-omic analyses of the human aortic arch adventitia identified PIEZO1, TRPV2, and TRPM4 as candidate mechanosensitive proteins associated with baroreceptor function, whereas TMEM150C was not among the most enriched candidates in these data sets.³⁰ Collectively, available human data do not support TMEM150C as a leading baroreceptor candidate.

Parallel With Other Mechanosensory Systems

A comparison can be drawn between arterial baroreceptors and muscle spindle afferents, which appear to share a convergent molecular logic centered on mechanically gated cation channels. In the muscle spindle, PIEZO2 is essential for converting the stretch of intrafusal fibers into receptor potentials in Group Ia/II afferents; its deletion nearly abolishes stretch-evoked firing, underscoring its role as the principal mechanotransducer.^{120,121} Another parallel lies in the mismatch between the rapid intrinsic kinetics of PIEZO2 currents and the slowly adapting, sustained firing patterns observed physiologically. In muscle spindles, this discrepancy is resolved through accessory mechanisms, including cytoskeletal coupling, lipid microdomains, and the contribution of additional mechanically sensitive channels.¹²² In particular, TMEM150c, as well as members of the DEG/ENaC and TRP channel families, are thought to complement PIEZO2 and help sustain

depolarizing currents during static stretch.^{115,123,124} In both systems, the architectural specialization of sensory endings, annulospiral endings in spindles versus end-net and flower-spray morphologies in baroreceptors, likely tunes the spatial distribution of mechanical stress and thereby modulates channel activation thresholds. This has been elegantly established in muscle spindle afferents, where mechanosensory encoding emerges from the interplay between neuronal architecture and the distribution and complement of ion channels, which together act as complementary determinants of sensory responsiveness.¹²⁵ This structural-functional coupling supports the idea that mechanosensitivity is not solely an intrinsic property of ion channels but also an emergent feature of the sensory ending's geometry and its interaction with surrounding tissues.¹²² This functional ensemble is further supported by nonchannel amplification mechanisms. Notably, muscle spindle afferents employ vesicular glutamate release to sustain firing during the static phase of stretch, thereby enhancing excitability downstream of the initial mechanotransduction event.^{126–128} A closely related paradigm is provided by Merkel cells in cutaneous touch receptors, where non-neuronal epithelial cells actively contribute to afferent firing.¹²⁹ In these complexes, Merkel cells release neurotransmitters, primarily serotonin, onto associated A β afferents, thereby driving and sustaining SAI (slowly adapting type I) firing. Such parallels suggest that sustained firing in mechanosensory systems is not solely dependent on ion channel kinetics but emerges from hybrid transduction units combining mechanosensitive channels with synaptic-like signaling. This concept may extend to baroreceptor endings that are equipped with the machinery necessary for the release of neuromodulators¹³⁰ (see Figure 2). Preliminary functional evidence indicates that glutamate can be released locally at baroreceptor endings and act on "autoreceptors" to modulate baroreceptor response.¹³¹

These findings support a general role for glutamate in the modulation of peripheral mechanosensory terminals, placing baroreceptors within a broader class of sensory endings that employ "efferent-like" chemical signaling. However, further studies are required to clarify the precise functional significance of this peripheral transmitter release.

CONCLUDING PERSPECTIVES

Baroreceptors have long been regarded as faithful sentinels of cardiovascular stability. Yet the molecular language through which they convert vascular stretch into neural information is only now revealing its syntax and complexity. What once appeared to be a unitary reflex mediated by a single class of mechanosensor is increasingly understood as a distributed sensory system composed of multiple ion channels, structurally distinct nerve endings, and dynamically regulated firing programs.

Rather than reflecting molecular monotheism, baroreception emerges as an ensemble process in which parallel mechanosensitive pathways encode complementary features of hemodynamic force. This plural architecture likely underpins the resilience, redundancy, and adaptability required for lifelong blood pressure regulation.

Future progress in baroreception will likely depend on integrating single-cell transcriptomics with functional mapping to resolve which vagal and glossopharyngeal afferents are true baroreceptors. Building on data sets from Kupari et al¹⁸² and Jo et al,¹³² baroreceptive neurons are predicted to form a distinct transcriptional cluster defined by coexpression of key mechanotransducers and excitability genes, including Piezo2, Asic2, Scn10a (Nav1.8), and Scn11a (Nav1.9), together with A-fiber markers such as Nefh and Pvalb, consistent with rapid and sustained mechanosensory signaling. A minimal genetic signature may further include P2rx2/3 and high VGLUT2 with low CGRP, distinguishing these neurons from nociceptive afferents. Phenotypes associated with hypertensive states may display reduced mechanosensitivity (lower Piezo2), increased expression of inhibitory potassium channels (eg, Kv1), enhanced Kcnq2/3-mediated stabilization, and possibly elevated Kcc2, contributing to diminished excitability. These multidimensional molecular fingerprints suggest that baroreceptors are defined by combinatorial gene expression rather than single markers, enabling intersectional genetic targeting. They also provide a conceptual framework for bidirectional therapeutic modulation, where enhancing mechanotransduction or excitability (eg, via Piezo2 gain-of-function or Kcnq inhibition) could restore depressor signaling, while reducing excitability (eg, via Kcnq activation or Nav1.8 inhibition) may counteract hypotensive states.

These advances shift the central questions of the field from identification to integration. Do native baroreceptor endings generate mechanically activated currents under physiological strain in intact vessels? How are forces transmitted through the remodeled or stiffened vascular wall to reach sensory terminals? What roles are played by endothelial cells, glial elements, extracellular matrix composition, and vascular biomechanics in shaping afferent signaling? Clarifying the functional specialization of distinct terminal morphologies and determining how adult-onset perturbations avoid developmental compensation observed in genetic models will be essential for linking molecular identity to actual firing behavior. Bridging transcriptomic signatures with in situ electrophysiology remains a critical step toward functional validation.

Biological variables further expand this complexity. Sex- and age-dependent transcriptional remodeling within vagal sensory pathways indicates that baroreceptor function is dynamically tuned across the lifespan.¹³² In parallel, metabolic, inflammatory, and circadian signals modulate mechanosensory excitability, providing a

plausible framework for the context-dependent failure of baroreflex buffering observed in cardiometabolic disease. Baroreception should therefore be viewed not as a static reflex, but as a sensorimotor system embedded within whole-body physiology.

These conceptual shifts compel a reassessment of cardiovascular pathology. In hypertension, heart failure, and syndromes characterized by heightened blood pressure variability, baroreceptor dysfunction may not simply reflect a displaced operating point along a fixed curve. Instead, selective impairment of specific sensory modules, through vascular remodeling, altered force transmission, metabolic stress, or channel dysregulation, may degrade the diversity of afferent encoding and weaken reflex buffering. Disease may thus represent not only altered set points but a reorganization of the sensory architecture itself.

Clinically, the partial success of bioelectronic baroreflex activation therapies demonstrates that peripheral mechanosensory circuits remain plastic and therapeutically accessible even in advanced disease. However, the variability of patient responses underscores the need for molecular, functional, or imaging biomarkers capable of defining barosensory integrity and stratifying therapeutic responsiveness. A mechanistic understanding of baroreceptor diversity could enable targeted pharmacological, genetic, or neuromodulatory strategies aimed at restoring physiological reflex homeostasis rather than merely compensating for its loss.

The next frontier in baroreceptor research will therefore be explicitly integrative. Progress will depend on uniting high-resolution anatomy, vascular biomechanics, electrophysiology in intact preparations, single-cell and spatial omics, and computational modeling capable of linking molecular events to systems-level reflex control. Such synthesis promises to elevate baroreception from its classical status as a canonical reflex to a broader paradigm of sensory biology, 1 in which mechanical, electrical, and molecular logics converge to ensure long-term cardiovascular stability. By transforming mechanistic diversity from a source of controversy into a framework for precision physiology, this approach may ultimately open new avenues for individualized control of cardiovascular homeostasis.

ARTICLE INFORMATION

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